

Notes: Borisova & Megginson (RFS, 2011)

Does Government Ownership Affect the Cost of Debt? Evidence from Privatization

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1 What this paper asks

This paper asks whether *retained government ownership* affects a privatized firm's *cost of debt*, measured by corporate bond credit spreads. The central empirical object is the mapping

$$\text{state ownership} \longrightarrow \text{bondholders' required return.}$$

The paper's key empirical finding is *non-monotonicity*: decreasing government ownership is associated with *higher* credit spreads (weaker implicit guarantees), yet *fully privatized* firms exhibit *lower* spreads than *partially privatized* firms (reduced ownership uncertainty and stakeholder conflict once the state completely exits). The net pattern is consistent with spreads being *lowest at the extremes* (near-zero or very high state ownership) and *highest at intermediate ownership*.

2 Economic mechanisms and testable predictions

The paper is best read as testing multiple channels that can operate in opposite directions.

2.1 Implicit guarantee channel (state as backstop)

- **Idea.** If the government retains ownership, bond investors may price an implicit repayment guarantee.
- **Prediction.** Higher retained government ownership $\Rightarrow$ lower credit spreads.
- **Crisis amplification.** The guarantee is more valuable in financial stress; the ownership–spread slope should become more negative during crisis years.

2.2 Efficiency / performance channel (privatization improves operations)

- **Idea.** Privatization can increase efficiency, profitability, and governance quality, lowering default risk.
- **Prediction.** Moving toward private ownership can lower spreads *once* privatization is complete and governance is stabilized.

2.3 Ownership uncertainty and stakeholder conflict (mid-privatization wedge)

- **Idea.** During gradual privatization, future tranche sales and unclear control rights can increase volatility and conflict between new shareholders and existing debtholders.
- **Prediction.** Partially privatized firms (especially with mid-range state stakes) may have higher spreads than fully privatized firms, even controlling for the percentage stake.

2.4 Golden shares as a “role-clarifying” governance device

- **Idea.** A golden share grants special state control rights; it is often viewed as interference, but it may also clarify the state’s role and stabilize expectations.
- **Prediction.** Golden shares may reduce spreads *conditional on partial state ownership* if they reduce uncertainty about the state’s control/commitment.

3 Data and key variables

3.1 Sample construction

- Unit of observation: bond–year (or bond–issue for at-issue specifications).
- Main sample: 1,651 bond-year observations from 60 European privatized companies over 2001–2009 (Table 1).
- The paper reports results for the full sample, EMU subsample, and non-bank subsample.

3.2 Dependent variable: credit spread

Let y_{it} denote the credit spread for bond i in year t :

$$y_{it} \equiv \text{YTM}_{it} - \text{YTM}_{m(it),t}^{gov}, \quad (1)$$

where $\text{YTM}_{m(it),t}^{gov}$ is the yield on the government bond most closely matched by maturity to bond i in year t .

3.3 Key regressors

- $\text{GovtOwnership}_{it}$: percent of shares still owned by the government.
- $\mathbb{1}\{\text{PartiallyPrivatized}_{it}\}$: indicator equal to 1 if the government retains any ownership.
- $\mathbb{1}\{\text{GoldenShare}_i\}$: indicator for special control rights retained by the state.
- Speed-of-privatization variables (Table 6):
 - LengthInitial : years since initial privatization.
 - LengthFully : years it took to become fully privatized.
 - $\mathbb{1}\{\text{ImmediatePrivatization}\}$: equals 1 if full privatization occurred in one transaction.

3.4 Controls

The controls include bond characteristics (age, maturity, currency, collateral/instrument dummies) and firm characteristics (market-to-book, size, ROE, leverage, bank dummy, and interactions), plus country-level characteristics (EMU membership, sovereign rating, creditor rights in some specifications). Table 2 lists definitions and predicted signs.

4 Empirical models (all specifications used in the paper)

4.1 Baseline panel regression (Eq. 1 in the paper)

The baseline model is:

$$y_{it} = \alpha + X'_{it}\beta + \gamma r_{it} + v_t + \varepsilon_{it}, \quad (2)$$

where:

- y_{it} is the credit spread,
- X_{it} contains the ownership variables and controls,
- r_{it} is the bond credit rating,
- v_t are year fixed effects,
- ε_{it} is the error term.

Inference uses heteroskedasticity-robust standard errors, clustered by firm, consistent with the paper's discussion.

4.2 Orthogonalization of credit ratings

Because credit ratings are themselves shaped by observables in X_{it} , the paper uses a residual transformation (orthogonalization) so that the rating control captures rating variation not mechanically explained by other regressors. Conceptually:

$$r_{it} = X'_{it}\pi + u_{it}, \quad (3)$$

$$\hat{\zeta}_{it} \equiv \hat{u}_{it}, \quad (4)$$

$$y_{it} = \alpha + X'_{it}\beta + \gamma \hat{\zeta}_{it} + v_t + \varepsilon_{it}. \quad (5)$$

This procedure is meant to reduce multicollinearity and clarify interpretation of the ownership coefficients conditional on rating.

4.3 Nonlinearity via “presence” plus “percentage”

A core empirical design feature is including both: (i) the *level* of ownership (GovtOwnership) and (ii) the *presence* indicator ($\mathbb{1}\{\text{PartiallyPrivatized}\}$). This yields a piecewise-linear shape:

$$\Delta y(\text{GovtOwnership} = x) \approx \delta \cdot \mathbb{1}\{x > 0\} + \beta_g x. \quad (6)$$

With $\delta > 0$ and $\beta_g < 0$, spreads are higher at low-but-positive x (partial privatization wedge), but decrease when x becomes large (implicit guarantee dominates).

4.4 Crisis interaction (guarantee more valuable in stress)

To test crisis amplification:

$$y_{it} = \alpha + \beta_g \text{GovtOwnership}_{it} + \beta_c (\text{GovtOwnership}_{it} \times \mathbb{I}\{t \in 2008, 2009\}) + \dots + v_t + \varepsilon_{it}. \quad (7)$$

A negative β_c supports a stronger guarantee channel during crisis.

4.5 Golden share interaction (role-clarification under partial ownership)

To test whether golden shares matter differently when the state remains an owner:

$$y_{it} = \alpha + \dots + \theta_1 \mathbb{I}\{\text{GoldenShare}\} + \theta_2 (\mathbb{I}\{\text{GoldenShare}\} \times \mathbb{I}\{\text{PartiallyPrivatized}\}) + v_t + \varepsilon_{it}. \quad (8)$$

4.6 Speed-of-privatization regressions (Table 6)

The paper examines whether the *pace* of privatization affects spreads:

$$y_{it} = \alpha + \beta_g \text{GovtOwnership}_{it} + \delta \mathbb{I}\{\text{PartiallyPrivatized}\} + \phi \text{LengthInitial}_{it} + \dots + v_t + \varepsilon_{it} \quad (\text{partial firms}), \quad (9)$$

$$y_{it} = \alpha + \psi \text{LengthFully}_i + \kappa \mathbb{I}\{\text{ImmediatePrivatization}_i\} + \dots + v_t + \varepsilon_{it} \quad (\text{fully privatized firms}). \quad (10)$$

4.7 Endogeneity and 2SLS instrumentation (Table 7)

Because government ownership may be endogenous (e.g., governments retain stakes in systematically safer or riskier firms), the paper uses 2SLS with excluded instruments Z_{it} . First stage (schematic):

$$\text{GovtOwnership}_{it} = Z'_{it} \lambda + X'_{it} \pi + v_t + \nu_{it}, \quad (11)$$

$$\mathbb{I}\{\text{PartiallyPrivatized}_{it}\} = Z'_{it} \lambda' + X'_{it} \pi' + v_t + \nu'_{it}, \quad (12)$$

Second stage:

$$y_{it} = \alpha + \beta_g \widehat{\text{GovtOwnership}}_{it} + \delta \widehat{\mathbb{I}\{\text{PartiallyPrivatized}\}}_{it} + X'_{it} \beta + \gamma \widehat{\zeta}_{it} + v_t + \varepsilon_{it}. \quad (13)$$

The excluded instruments emphasized in the paper include LengthInitial , the government's fiscal balance (as % of GDP), firm size proxies (e.g., employees), and privatization-wave timing dummies.

4.8 Operating uncertainty and ownership quartiles (Table 7, Models 5–6)

To study whether uncertainty is especially costly in mid-privatization:

$$y_{it} = \alpha + \sum_{q=1}^4 [\omega_q U_{it} \times \mathbb{I}\{\text{StakeQuartile} = q\}] + \text{main effects} + v_t + \varepsilon_{it}, \quad (14)$$

where U_{it} is either (i) $\text{StDev}(\text{ROA})$ or (ii) a high-uncertainty indicator. A positive interaction for intermediate quartiles supports the “uncertainty wedge” channel.

4.9 Quadratic ownership model (Table 8; Figures 3 and 4)

An alternative way to model nonlinearity is:

$$y_{it} = \alpha + b_1 \text{GovtOwnership}_{it} + b_2 \text{GovtOwnership}_{it}^2 + \cdots + v_t + \varepsilon_{it}. \quad (15)$$

With $b_1 > 0$ and $b_2 < 0$, spreads are hump-shaped in ownership (highest at intermediate state stakes). The implied turning point is:

$$x^* = -\frac{b_1}{2b_2}. \quad (16)$$

4.10 At-issue specification (Table 8, Models 3–4)

Using spreads at issuance removes bond-age liquidity concerns and adds duration/convexity controls:

$$y_i^{\text{issue}} = \alpha + \beta_g \text{GovtOwnership}_i + \delta \mathcal{K}\{\text{PartiallyPrivatized}\}_i + \Gamma' W_i + \text{FE} + \varepsilon_i, \quad (17)$$

where W_i includes maturity, duration, convexity, and issue amount (scaled).

4.11 Firm-level total cost of debt (Eq. 2; Table 8, Models 5–6)

The paper also uses a firm-level after-tax cost-of-debt measure:

$$\text{CoD}_{it} \equiv \left(\frac{I_{it}}{TD_{it}} \right) (1 - TR_{it}), \quad (18)$$

where I is interest expense, TD total debt, and TR the tax rate. Regressions use firm rating and industry fixed effects.

4.12 Within-firm ownership variation with firm fixed effects (Table 9, Model 1; Figure 4)

Restricting to observations where state ownership changes and adding firm fixed effects:

$$y_{it} = \alpha_i + b_1 \text{GovtOwnership}_{it} + b_2 \text{GovtOwnership}_{it}^2 + v_t + \varepsilon_{it}. \quad (19)$$

This isolates *within-firm* variation in ownership.

5 Read each table and figure

This section is organized in the order the paper presents results.

5.1 Table 1: Sample description

- The sample contains 60 privatized companies from 14 European countries and 1,651 bond-year observations (2001–2009).
- Observations are concentrated in large privatization programs (notably France) and include both banks and nonbanks.
- The strong country and industry clustering is a reason the paper emphasizes fixed effects and clustered standard errors later.

Table 1
Sample description

Country	Number of Companies			Number of Observations									
	Total	Banks	Nonbanks	Total	Banks	Nonbanks	2001	2002	2003	2004	2005	2006	2007
Austria	2	1	1	66	34	32	2	2	2	2	2	2	2
Belgium	2	1	1	14	0	14	1	1	1	1	1	1	1
Denmark	1	0	1	14	0	14	1	1	1	1	1	1	1
France	4	0	4	461	0	461	40	3	4	7	8	12	2
Germany	17	6	11	754	421	293	69	82	101	119	119	82	40
Greece	4	2	2	240	101	139	1	19	21	31	42	40	2
Ireland	1	0	1	8	0	8	1	1	1	1	1	1	1
Italy	2	1	1	65	30	35	46	18	21	21	10	10	6
Netherlands	2	0	2	226	101	125	19	21	21	10	10	10	2
Portugal	1	0	1	45	0	45	1	1	1	1	1	1	1
Spain	2	1	1	46	0	46	1	1	1	1	1	1	1
Sweden	1	0	1	36	0	36	1	1	1	1	1	1	1
UK	11	0	11	1,000	0	1,000	11	11	11	11	11	11	11
Total	49	17	32	2,451	622	1,829	241	110	241	280	302	240	64

Real data are collected from Bloomberg for bond ratings (provided from Bond) in the Privatization Barometer database. Firm data are collected from the Thomson ONE Banker database. Real data cover the period 2001–2009 and are constructed by the availability of the Bloomberg credit spread data.

Table 2
Continued

Variable	Definition
Bank Guaranteed	A security whose payment of principal and interest are guaranteed by a bank
Bonds	A certificate of debt issued by a government or corporation with the promise to pay back the principal amount as well as interest by a specified future date
Collateralized	Securities that are secured by an asset of the issuer, frequently a mortgage
Company Guant	A bond whose principal and interest payments are guaranteed by an outside company
Covered	A covered or mortgage bond is an on-balance sheet obligation of the institution. Typically AAA rated, a covered bond receives the legal structure, the issuer's backing, and the pledge of quality assets
Local Govt Guant	Local government guarantee. A bond whose principal and interest payments are guaranteed by a local government, such as a province, city, or town
Notes	A legal document that is evidence of a debt and that requires payment within a specific time period
Pfandbrief	Legally protected term for bonds issued by German banks, subject to special Pfandbrief legislation
Secured	A bond for which the issuer has set aside assets as collateral to ensure timely interest and principal payments
Senior Notes	A security that has priority over other securities in claiming assets and dividends
Sr Sub Notes	Senior subordinated notes. Ranks lower in priority than senior debt when dividends are paid or assets are distributed in liquidation or bankruptcy, but ranks higher than junior subordinated debt
Sr Subordinated	Senior subordinated. Ranks lower in priority than senior debt when dividends are paid or assets are distributed in liquidation or bankruptcy, but ranks higher than junior subordinated debt
Sr Unsecured	Senior unsecured. A security that has priority over other bonds in claiming assets and dividends
Sr Unsub	A note with a lower priority than other securities when dividends are paid or assets are distributed in liquidation or bankruptcy
Sub Notes	A security with a lower priority than other securities when dividends are paid or assets are distributed in liquidation or bankruptcy
Subordinated	A debt obligation with no collateral, and backed only by the debtor's creditworthiness
Unsecured	A debt obligation with no collateral, and backed only by the debtor's creditworthiness
Unsubordinated	A security that does not rank lower than other securities when dividends are paid or assets are distributed in liquidation or bankruptcy

Privatization variables are from Privatization Barometer. Bond data are obtained from Bloomberg. Financial and block ownership data are obtained from the Thomson ONE Banker database. The variable *Creditor rights* is from Andrei Shleifer's website. All monetary units are specified in euros. Whenever data are in any other currency, conversion to euros has been performed using the exchange rate as of the end of the fiscal year of the respective company. Exchange rates have been obtained using historical data from the European Central Bank website (www.ecb.int) and Bloomberg. Predicted signs of the regressors' influence on *Credit spread* are shown in the right-hand column.

Table 2
Variable definitions and predicted relations with the cost of debt

Variable	Definition	Prediction
Credit spread	The difference between the current yield to maturity of the corporate bond under review and that of a government bond of the respective country, in basis points	
Govt ownership	Percentage of the company still owned by the government. Obtained from www.privatizationbarometer.net and company financial statements	(+) / (-)
Partially privatized	Takes a value of 1 if the company currently has some government ownership and 0 otherwise	(+) / (-)
Golden share	Takes a value of 1 if the government has a golden share, which grants it special powers, and 0 otherwise	(+) / (-)
Length fully	The number of years it took the government to completely divest the company (only applicable to fully privatized companies)	(+)
Length initial	The number of years since the company was privatized initially	(+)
Immediate privatization	Takes a value of 1 if the company was fully privatized in one transaction and 0 otherwise	(-)
Rating	The natural log of Moody's bond rating, after conversion to an ordinal scale (Aaa = 19, Aa1 = 18, etc.)	(-)
Country rating	The natural log of the country's credit rating provided by Moody's, after conversion to an ordinal scale (Aaa = 19, Aa1 = 18, etc.)	(+)
Age	The time since the issue date, in days	(+) / (-)
Maturity	The time till maturity, in days	(+)
Euro	Takes a value of 1 if the bond issue is euro denominated and 0 otherwise	(-)
Amount issued	The amount of the bond issue scaled by beginning of period assets	(-)
Eurobond	Takes a value of 1 if the bond is a Eurobond based on the market of issue from Bloomberg and 0 otherwise	(-)
Duration	Macaulay's duration of the bond at issue, obtained from Bloomberg	(+)
Convexity	The rate of change of a bond's duration as yields change, obtained from Bloomberg	(-)
Creditor rights	A country-level composite index of creditors' rights, ranging from 0 (weak) to 4 (strong) used by Djankov et al. (2008)	(-)
Leverage	(Total assets - Stockholders equity) / Stockholders equity	(+)
M_B	(Total shares * Closing share price) / Stockholders equity	(+) / (-)
Size	The natural log of total assets	(-)
ROE	Net income / Stockholders equity	(+)
ROS	Net income / Sales	(-)
Interest coverage	Earnings before interest and taxes / (Interest expense on debt + Preferred dividends)	(-)
Block ownership	Percentage of the company owned by blockholders (more than 5%) other than the government	(+)
Year after losing auditor	Takes a value of 1 if the company lost a previously employed auditor during the previous year and 0 otherwise	(+)
Bank	Takes a value of 1 if the company is a bank and 0 otherwise	(-)
MA	Takes a value of 1 if the company, following its initial privatization, has been involved in a merger or acquisition prior to the time of the bond spread measurement and 0 otherwise	(+) / (-)
EMU	Takes a value of 1 if the company is headquartered in the European Monetary Union and 0 otherwise	(-)
St. Dev. ROA	Firm-level standard deviation of (Net income / Total assets) * 100 over the 9-year period	(+)
High operating uncertainty	Takes a value of 1 if the standard deviation of a firm's ROA is above the median and 0 otherwise	(+)
Collateral and Instrument Types	Takes a value of 1 if the bond is classified as this collateral or instrument type by Bloomberg and 0 otherwise. There are 18 types described below. Definitions are from the Bloomberg System	

(continued)

5.2 Table 2: Variable definitions and predicted relations

- The table lists each variable, how it is constructed, and the expected sign on spreads.
- Key conceptual point: many signs are *a priori ambiguous* because privatization affects spreads through opposing channels (guarantee vs efficiency vs uncertainty).

5.3 Table 3: Descriptive statistics

- Mean credit spread is about 81 bps; the sample includes both investment-grade and lower-rated bonds.
- Average retained government ownership is about 10% across all observations, but conditional on being partially privatized, mean ownership is much higher (about 33%).
- The presence of golden shares and the significant bank share of observations motivate heterogeneity checks.

5.4 Table 4: Differences in means (fully vs partially privatized)

- Partially privatized firms have higher mean spreads than fully privatized firms in the pre-crisis period (2001–2007), consistent with a partial-privatization wedge.
- During 2008–2009, spreads rise sharply overall; differences between full and partial are weaker, consistent with crisis-related repricing and the guarantee channel becoming more salient.

5.5 Table 5: Baseline regressions — retained ownership and spreads

Table 5 is the core evidence for the paper.

- **Retained ownership effect:** GovtOwnership enters with a *negative* and highly significant coefficient across models (Model 1: about -0.81 bps per 1 percentage point of ownership).
 - Interpretation: higher state ownership lowers spreads, consistent with an implicit guarantee.
 - Economic magnitude: a 10pp higher state stake is associated with roughly 8 bps lower spreads in Model 1.

Table 3
Descriptive statistics

Continuous Variables	Observations	Mean	Standard Deviation	Min	Max
Credit spread	1,651	80.60	79.24	0.00	515.00
<i>Privatization variables</i>					
Govt ownership	1,651	10.33	20.61	0.00	87.30
Govt ownership > 0	516	33.05	24.66	2.91	87.30
Length fully	1,135	2.03	4.13	0.00	19.00
Length initial	1,651	12.09	5.63	0.00	28.00
<i>Bond variables</i>					
Rating	1,651	15.21	2.52	6.00	19.00
Age	1,651	1,869.11	1,132.22	13.00	6,035.00
Maturity	1,651	2,317.35	1,994.87	9.00	10,785.00
Amount issued	1,604	1.74	3.19	0.0004	19.37
Duration	1,645	8.32	2.40	1.94	16.00
Convexity	1,645	0.90	0.59	0.05	3.45
<i>Firm variables</i>					
Leverage	1,651	17.13	18.65	0.40	197.39
M_B	1,651	1.94	2.04	0.21	42.69
Size	1,651	11.93	1.57	7.55	14.89
ROE	1,651	0.12	0.18	-2.63	1.05
ROS	1,651	0.08	0.10	-1.54	0.59
Interest coverage	1,651	5.57	16.13	-69.90	234.33
St. Dev. ROA	1,651	1.62	3.25	0.12	31.76
Block ownership	1,651	27.73	29.40	0.00	96.35
<i>Country variables</i>					
Country rating	1,651	18.63	0.84	13.00	19.00
Creditor rights	1,651	1.49	1.46	0.00	4.00
Binary Variables	Observations		Yes (1)		No (0)
<i>Privatization variables</i>					
Partially privatized	1,651		516		1,135
Golden share	1,651		260		1,391
Immediate privatization	1,651		766		885
<i>Bond variables</i>					
Euro	1,651		1,046		605
Eurobond	1,651		934		717
<i>Firm variables</i>					
Bank	1,651		922		729
MA	1,651		797		854
EMU	1,651		1,461		190
Year after losing auditor	1,651		144		1,507

The table describes the number, mean, standard deviation, minimum, and maximum of the variables used in the analysis. Variable definitions are provided in Table 2. The sample covers the period 2001–2009. Credit spreads in the top and bottom 1% of all observations are dropped, and negative market-to-book values (1 firm-year) are dropped.

Table 4
Differences in means tests

Variable	Full Sample	Fully Privatized	Partially Privatized	P-value	Count
Credit spread	80.60	78.98	84.16	0.2180	1,651
Credit spread (01-07)	58.06	56.13	62.48	0.0275	1,336
Credit spread (08-09)	176.17	182.03	165.11	0.1866	315
<i>Privatization variables</i>					
Govt ownership	10.33	0.00	33.05	0.0000	1,651
Golden share	0.16	0.09	0.31	0.0000	1,651
Length initial	12.09	13.35	9.32	0.0000	1,651
<i>Bond variables</i>					
Rating	2.71	2.74	2.64	0.0000	1,651
Age	1,869.11	1,939.85	1,713.49	0.0002	1,651
Maturity	2,317.35	2,401.13	2,133.06	0.0113	1,651
Euro	0.63	0.60	0.72	0.0000	1,651
Eurobond	0.57	0.50	0.72	0.0000	1,651
Amount issued	1.74	1.42	2.45	0.0000	1,604
Duration	8.32	8.49	7.93	0.0000	1,645
Convexity	0.90	0.94	0.81	0.0000	1,645
<i>Firm variables</i>					
Leverage	17.13	19.43	12.07	0.0000	1,651
M_B	1.94	1.93	1.96	0.7717	1,651
Size	11.77	12.04	11.15	0.0000	1,651
ROE	0.12	0.13	0.10	0.0013	1,651
ROS	0.08	0.08	0.06	0.0000	1,651
Interest coverage	5.57	6.16	4.29	0.0294	1,651
St. Dev. ROA	1.62	1.30	2.33	0.0000	1,651
Block ownership	27.73	35.87	9.80	0.0000	1,651
Year after losing auditor	0.09	0.08	0.10	0.1326	1,651
Bank	0.56	0.71	0.22	0.0000	1,651
MA	0.48	0.57	0.29	0.0000	1,651
EMU	0.88	0.87	0.93	0.0004	1,651

The following table presents two-tailed tests of differences in means for fully vs. partially privatized companies for the dependent variable (*Credit spread*) and all independent variables. Variable definitions are provided in Table 2. The sample covers the period 2001–2009. A few variables are constrained by data availability. *P*-value shows the significance of the two-tailed difference in means test. Differences for collateral type, country, and currency are not reported but are available upon request.

- **Partial-privatization wedge:** the Partially privatized dummy is *positive* and significant (Model 1: about +25 bps).

- Interpretation: holding the ownership percentage fixed, simply being “mid-privatization” is costly for bond pricing.
- Mechanism: uncertainty about future stake sales and control rights, plus stakeholder conflict between new shareholders and debtholders.
- **Implied threshold (Model 1):** the discrete premium is offset once

$$x \gtrsim \frac{25}{0.81} \approx 31\%,$$

meaning that at *low* state stakes partial privatization is priced as risky, while at *high* state stakes the implicit guarantee dominates.

Table 5
Retained government ownership and the cost of debt

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Govt ownership	-0.811*** (-6.9981)	-0.758*** (-5.1847)	-0.851*** (-6.9859)	-0.851*** (-7.5231)	-0.662*** (-5.5337)	-0.751*** (-4.3948)
Govt ownership * Fin. crisis					-0.561*** (-2.8428)	
Partially privatized	24.84*** (3.0405)	32.30*** (5.2114)	18.46** (2.1545)	29.64*** (3.5797)	25.09*** (3.5262)	16.98* (1.7200)
Golden share	-8.79 (-1.4956)	-17.78*** (-3.0550)	-8.90 (-1.4893)	2.93 (0.4244)	-5.18 (-0.8425)	-10.61 (-1.0042)
Golden share * Partially privatized					-33.48*** (-3.1098)	
Rating	-181.9*** (-10.2545)	-156.6*** (-8.7671)	-204.2*** (-11.4809)	-179.1*** (-9.2018)	-183.5*** (-10.0376)	
Country rating					101.2** (2.2154)	
Age	-0.00251 (-1.2037)	-0.00514** (-2.9243)	0.000343 (0.1044)	-0.00233 (-1.1134)	-0.00208 (-1.3656)	
Maturity	0.000300 (0.3052)	0.000177 (0.1471)	0.00146 (1.3432)	0.000683 (0.6801)	0.000620 (0.6074)	
Euro	-13.11*** (-2.7178)	-12.60** (-2.4652)	-13.32* (-1.8085)	-13.00*** (-2.6954)	-11.32** (-2.4469)	
M, B	1.188 (0.7282)	-1.239 (-0.7591)	2.704 (1.4921)	0.857 (0.5143)	1.428 (0.8580)	0.397 (0.1902)
Size	-0.551 (-0.3338)	-1.033 (-4.5331)	-2.863 (-7.9133)	-0.442 (-0.2761)	-0.363 (-0.2350)	5.808 (1.5618)
ROE	-72.11*** (-5.6701)	-86.54*** (-4.7443)	-71.50*** (-5.6988)	-72.55*** (-5.5992)	-70.56*** (-5.6337)	
Bank	-20.74** (-2.1159)	-17.45** (-2.0224)	-19.80** (-2.0315)	-15.76* (-1.7960)	-22.28* (-1.9268)	
Leverage	0.110 (0.2975)	1.855 (0.8222)	-0.241 (-0.6017)	0.181 (0.4772)	0.0474 (0.1250)	0.195 (0.3934)
Bank * Leverage	-0.437 (-1.2053)	-2.277 (-1.0195)	-0.555 (-1.4955)	-0.415 (-1.1905)	-0.505 (-1.1905)	
MA	-15.98*** (-3.0775)	-6.892 (-1.2674)	-31.99*** (-3.8093)	-16.70*** (-3.1916)	-16.46*** (-3.4991)	-19.51** (-2.3842)
EMU	-49.32*** (-5.6145)	-46.88*** (-5.2542)	-46.88*** (-5.5303)	-48.42*** (-5.5303)	-48.42*** (-5.5303)	-58.21*** (-4.0396)

(continued)

Table 5
Continued

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Block ownership					0.0771 (0.5252)	
Interest coverage					-0.0807 (-0.6882)	
Constant	153.3*** (7.8903)	124.6*** (6.3137)	181.4*** (6.4372)	147.8*** (7.6635)	-149.2 (-1.0807)	2.4148
Collateral/instrument dummies	Yes	Yes	Yes	Yes	Yes	No
Observations	1651	1461	729	1651	1651	1651
R^2 -squared	0.3874	0.3001	0.4832	0.3880	0.4090	0.1910

Year fixed effects (ν_i) regression with heteroscedasticity-robust and firm-clustered standard errors is performed on the following model: $y_{it} = \alpha + \beta X_{it} + \gamma \tilde{L}_{it} + \nu_i + \eta_t$.
The dependent variable, *Credit spread* (y_{it}), is the difference between the corporate bond's current yield to maturity and that of a government bond that is most closely matched by maturity. α represents the intercept, and η_t is the error term. Orthogonalized values of the log of the bond's rating after conversion to an ordinal scale, *Rating* ($\tilde{L}_{it}$), are used. The explanatory variables included in X_{it} are described in Table 2. *Bank * Leverage* is the interaction of these two variables. Model 2 uses only firms from countries in the EMU. Model 3 includes only non-bank firms to control for the possibility that banking firms drive the results. Model 4 includes an interaction between golden share and the partially privatized indicator. Model 5 employs national credit ratings (*Country rating*) and an interaction between retained state ownership and a binary variable taking a value of 1 for observations from 2008 or 2009 (*Govt ownership * Fin. crisis*). Model 6 drops all bond-specific regressors. The data are annual and cover the period 2001–2009. The table reports estimated coefficients, with t -statistics in parentheses. *** denotes significance at the 1% level, ** denotes significance at the 5% level, and * denotes significance at the 10% level.

Table 6
Retained government ownership, the cost of debt, and the speed of privatization

Variable	Model 1	Model 2	Model 3	Model 4
Govt ownership	-0.525*** (-2.7044)			-0.780*** (-7.0073)
Partially privatized				15.75* (1.8419)
Golden share	-33.52*** (-5.1516)	1.149 (0.1687)	6.139 (0.7214)	-8.175 (-1.4432)
Length initial	1.755** (2.1711)			
Length fully		3.840*** (7.5313)		
Immediate privatization				-27.45*** (-3.4323)
Rating	-182.2*** (-6.0263)	-175.0*** (-5.6320)		-181.3*** (-6.1452)
Age	0.00169 (0.4512)	-0.00576*** (-2.7282)	-0.00508** (-2.3364)	-0.00260 (-1.3274)
Maturity	0.00512*** (3.4069)	-0.00110 (-0.9174)	-0.00150 (-1.2442)	0.000366 (0.3824)
Euro	-13.44* (-2.0110)	-13.93** (-2.1938)	-13.65** (-2.1500)	-12.16** (-2.5882)
M, B	-11.67** (-2.1121)	0.862 (0.4713)	0.866 (0.4625)	0.837 (0.4841)
Size	-8.359** (-2.0674)	1.190 (0.7751)	5.545*** (3.2322)	1.077 (0.6340)
ROE	-57.40*** (-4.0145)	-57.40*** (-4.5554)	-55.19*** (-4.3176)	-73.49*** (-5.8128)
Bank	-27.27 (-1.1143)	7.751 (0.6155)	-13.36 (-0.9388)	-14.75 (-1.2765)
Leverage	6.357* (2.0447)	0.0365 (0.0932)	0.0638 (0.1589)	0.116 (0.3032)
Bank * Leverage	-6.681** (-2.1532)	-0.342 (-0.5920)	-0.208 (-0.3340)	-0.445 (-1.1626)
MA	12.36 (0.9356)	-16.92*** (-3.4192)	-19.40*** (-3.9544)	-17.96*** (-3.4174)
EMU	-0.920 (-0.0590)	-89.33*** (-9.3804)	-84.59*** (-7.8636)	-59.28*** (-6.0056)
Constant	193.8*** (4.0909)	149.1*** (7.5087)	131.7*** (5.8406)	154.8*** (7.8561)
Collateral/instrument dummies	Yes	Yes	Yes	Yes
Observations	516	1135	1135	1651
R^2 -squared	0.3473	0.4621	0.4565	0.3902

Year fixed effects (ν_i) regression with heteroscedasticity-robust and firm-clustered standard errors is performed on the following model: $y_{it} = \alpha + \beta X_{it} + \gamma \tilde{L}_{it} + \nu_i + \eta_t$.
The dependent variable, *Credit spread* (y_{it}), is the difference between the corporate bond's current yield to maturity and that of a government bond that is most closely matched by maturity. α represents the intercept, and η_t is the error term. Orthogonalized values of the log of the bond's rating after conversion to an ordinal scale, *Rating* ($\tilde{L}_{it}$), are used. The explanatory variables included in X_{it} are described in Table 2. *Bank * Leverage* is the interaction of these two variables. Model 1 includes only non-bank firms and a variable representing immediate privatization.

- **Crisis amplification:** in Model 5, $\text{GovtOwnership} \times \mathbb{I}\{2008, 2009\}$ is negative and significant.

- Interpretation: implicit guarantees are valued more during stress; the ownership–spread slope becomes more negative.

- **Golden shares:** Model 4 shows the interaction $\text{GoldenShare} \times \text{PartiallyPrivatized}$ is negative and significant.

- Interpretation: when the state remains an owner, a golden share may reduce uncertainty by clarifying the state's role and commitment.

5.6 Figure 1: Raw relationship between ownership and spreads

Figure 1 plots average spreads against government ownership levels (a “nonparametric” first look).

- Key visual takeaway: spreads do not fall smoothly as ownership changes; instead, intermediate ownership levels are associated with relatively high spreads, consistent with non-monotonicity.

5.7 Table 6: Speed of privatization

Table 6 adds timing variables to connect ownership structure with the privatization *process*.

- **Partial firms:** longer time since initial privatization (LengthInitial) is associated with higher spreads.
 - Interpretation: being “stuck” in partial privatization for longer is costly for bondholders.
- **Fully privatized firms:** longer time to full privatization (LengthFully) is strongly positively associated with spreads.
 - Interpretation: gradual privatization correlates with higher debt costs even after full exit, consistent with persistent uncertainty or costly transition.
- **Immediate full privatization:** $\nexists \{\text{ImmediatePrivatization}\}$ is negative and significant.
 - Interpretation: quickly resolving ownership uncertainty is associated with cheaper debt.

5.8 Table 7: Endogeneity, fixed effects, and uncertainty mechanisms

Table 7 strengthens identification and probes mechanisms.

- **2SLS (Models 1–2):** instrumenting ownership produces a similarly negative ownership coefficient (in magnitude often larger than OLS).
 - Interpretation: endogeneity concerns do not overturn the main guarantee result; if anything, OLS may understate the magnitude.
- **Fixed effects (Models 3–4):** currency and country FE preserve the key coefficients.
- **Uncertainty interactions (Models 5–6):** operating uncertainty raises spreads most strongly in intermediate ownership quartiles (especially the 50–75% range).
 - Interpretation: uncertainty is particularly costly when control is contested or future ownership sales are salient.
- **Information risk proxy:** spreads increase in the year after losing an auditor.
 - Interpretation: consistent with bondholder concerns about information quality and agency conflicts.

Table 6
Instrumented retained government ownership, fixed effects, operating uncertainty, and the cost of debt

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Term structure	-0.017*** (-1.7398)	-0.017*** (-1.7398)	-0.017*** (-1.7398)	-0.017*** (-1.7398)	-0.017*** (-1.7398)	-0.017*** (-1.7398)
Partially privatized	-0.002** (-2.0198)	-0.002** (-2.0198)	-0.002** (-2.0198)	-0.002** (-2.0198)	-0.002** (-2.0198)	-0.002** (-2.0198)
Golden rules	-0.001*** (-3.0482)	-0.001*** (-3.0482)	-0.001*** (-3.0482)	-0.001*** (-3.0482)	-0.001*** (-3.0482)	-0.001*** (-3.0482)
Rating	-0.002*** (-2.0198)	-0.002*** (-2.0198)	-0.002*** (-2.0198)	-0.002*** (-2.0198)	-0.002*** (-2.0198)	-0.002*** (-2.0198)
Age	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Size	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year after losing auditor	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Constant	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
Observations	100	100	100	100	100	100
R-squared	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

Table 7
Continued

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Age	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Size	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year after losing auditor	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Constant	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
Observations	100	100	100	100	100	100
R-squared	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

Table 8
Continued

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Age	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Size	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Year after losing auditor	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)	-0.0001 (-0.0001)
Constant	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
Observations	100	100	100	100	100	100
R-squared	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

5.9 Table 8 and Figures 2–3: Alternative specifications

- **Quadratic ownership model (Models 1–2; Figure 3):** for EMU firms, the fitted relation is hump-shaped. Using Table 8 Model 2, the turning point is approximately

$$x^* \approx \frac{0.805}{2 \times 0.0156} \approx 26\%,$$

so spreads are highest near intermediate state stakes and lower near the extremes.

- **At-issue spreads (Models 3–4):** results persist when focusing on issuance dates and using duration/convexity controls, reducing concerns about liquidity measurement using bond age.
- **Firm-level cost of debt (Models 5–6):** the negative relation between state ownership and debt costs carries over to a broader debt measure, not only bonds.
- **Figure 2:** visualizes the “discrete drop” in fitted spreads when the state fully exits, corresponding to the Partially privatized nonlinearity.

Table 8
Alternative specifications: A quadratic form of estimated government ownership, credit spreads at issue, and a firm-level cost of debt measure

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Ownership	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Ownership ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Partially privatized						
Issuance date	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Issuance date ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Firm rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Age	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Size	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Debt	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Constant	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
R-squared	0.0002	0.0007	0.0007	0.0007	0.0009	0.0005

Table 9
Continued

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Ownership	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Ownership ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Partially privatized						
Issuance date	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Issuance date ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Firm rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Age	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Size	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Debt	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Constant	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
R-squared	0.0002	0.0007	0.0007	0.0007	0.0009	0.0005

Table 9
Robustness checks: Within-firm ownership variation, additional fixed controls, and robustness based on FE

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Ownership	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Ownership ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Partially privatized						
Issuance date	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Issuance date ²	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
Rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Firm rating	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Age	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Size	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Debt	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
Constant	-0.0002 (0.0002)	0.0007 (0.0005)	-0.0007 (0.0007)	-0.0004 (0.0005)	-0.0009 (0.0009)	-0.0004 (0.0005)
R-squared	0.0002	0.0007	0.0007	0.0007	0.0009	0.0005

5.10 Table 9 and Figure 4: Robustness and heterogeneity

Table 9 complements the main analysis in two ways.

- **Within-firm changes (Model 1; Figure 4):** using only ownership-changing observations and firm FE yields a hump-shaped fitted relation. Using Table 9 Model 1,

$$x^* \approx \frac{5.490}{2 \times 0.0584} \approx 47\%,$$

so even within the same firm, intermediate state stakes are associated with the highest spreads.

- **Additional controls (Model 2):** adding interest coverage, block ownership, profitability (ROS), and creditor rights leaves core ownership results intact.
- **Risk subsample (Model 3):** the negative ownership effect is stronger among lower-rated bonds.
- **Sovereign creditworthiness split (Models 4–5):** results differ across high- vs lower-rated sovereigns:
 - In top-rated sovereigns, state ownership reduces spreads strongly.

- In lower-rated sovereigns, state ownership can raise spreads, consistent with “the guarantor is risky” and/or stronger political interference.
- **Extra bond controls (Model 6):** adding issue amount and Eurobond indicator does not overturn the main coefficients.

6 Short summary

Putting the evidence together, the paper supports the following integrated view:

1. **Guarantees matter for bond pricing:** when the state remains a large owner, spreads are lower, especially in crises.
2. **“Being in transition” is costly:** partial privatization is associated with a distinct spread premium, consistent with uncertainty and stakeholder conflict.
3. **Complete exit resolves the wedge:** spreads drop for fully privatized firms relative to partial ones, consistent with stabilization of control rights and improved efficiency.
4. **Institutions and sovereign risk condition everything:** the value of a guarantee depends on the guarantor’s own creditworthiness; in weaker sovereigns, state ownership may not help (and may hurt).

7 Conclusion and intuition

7.1 Summary

This paper asks whether *retained government ownership* affects a privatized firm’s *cost of debt*. Using a European sample of privatized former SOEs from the Privatization Barometer matched to bond data, the paper builds a bond-year panel of **1,651 bond-year observations** covering **308 bonds** issued by **60 firms** (from **14 countries** and **6 industries**) over **2001–2009**. Credit spreads are measured in basis points as the difference between a corporate bond’s current yield-to-maturity and a maturity-matched sovereign benchmark in the same currency (Bloomberg mid quotes).

The main conclusion is **non-monotonicity in the debt-market consequences of privatization**. As governments reduce their equity stakes, credit spreads **rise on average**, consistent with the loss of an implicit state guarantee. However, firms that are **fully privatized** have **lower spreads** than firms that remain **partially privatized**, indicating that a *lengthy or incomplete* privatization process is itself costly for debtholders. The evidence is consistent with multiple mechanisms operating at different stages of privatization: (i) diminishing government guarantees, (ii) post-privatization efficiency and competitiveness gains, and (iii) intermediate-stage frictions such as ownership uncertainty and bondholder–shareholder conflicts.

7.2 Main empirical conclusions (portable takeaways with anchors)

1. **Retained government ownership lowers credit spreads (implicit guarantee).** In the baseline panel regressions (Table 5), the coefficient on retained government ownership is negative and highly significant (e.g., about -0.758 to -0.851 in basis points per percentage point). A useful summary is:

$$\Delta \text{Spread} \approx +0.75 \text{ bp for a 1 percentage-point decrease in government ownership.}$$

This is consistent with debtholders pricing a weaker expected repayment backstop as the state exits.

2. **The guarantee effect is stronger in stress periods.** The interaction of government ownership with the 2008–2009 financial-crisis indicator is negative (Table 5), implying that the spread-reducing role of state backing becomes more valuable when default risk is higher.
3. **Partially privatized firms face an additional spread premium relative to fully privatized firms.** Conditional on controls, a *Partially privatized* indicator is positive and economically large (Table 5), on the order of $\sim 18\text{--}32$ **bp** depending on specification (the paper highlights an average effect around ~ 24 **bp** in full-sample models). This premium is above and beyond what is captured by the continuous government-ownership share itself.
4. **Speed of privatization matters: slower divestiture raises spreads.** Table 6 shows that longer privatization processes are associated with higher spreads:
 - For partially privatized firms, more years since initial privatization increase spreads (e.g., about $+1.76$ **bp per year**).
 - For fully privatized firms, taking longer to complete the divestiture increases spreads (e.g., about $+3.88$ **bp per year**).
 - “Immediate” privatization (rapid completion) is associated with **lower spreads** (about -19 to -27 **bp** depending on specification).
5. **Golden shares can reduce spreads when residual state ownership exists.** While golden shares are typically associated with potential state interference, the interaction evidence suggests they can *lower* debt costs for partially privatized firms by clarifying the government’s role and the boundary of state control (e.g., a large negative interaction effect in Table 5).

7.3 Economic intuition: why the pattern is non-monotone

A coherent way to reconcile the findings is to view privatization as moving firms across three regimes:

1. **High state ownership (pre- or early privatization): low spreads via guarantee.** When the government retains a large stake, debtholders price an implicit repayment guarantee. This lowers spreads, especially in downturns when guarantees are most valuable.

2. **Intermediate state ownership (partial privatization): high spreads via uncertainty and stakeholder conflict.** As the state starts selling down, the guarantee weakens, but the firm is not yet fully governed by private-market discipline. This intermediate phase can raise spreads because:

- *ownership uncertainty* increases (future tranches, political timing, and control rights remain unclear),
- *volatility perceptions* rise due to restructuring and policy uncertainty,
- *bondholder–shareholder conflict* can intensify as new equityholders push for strategies that transfer value from debtholders (or as governments attempt to maximize tranche-sale revenues by catering to equity at debtholders’ expense).

These forces generate the observed “partial-privatization premium” in spreads.

3. **Zero state ownership (full privatization): spreads fall as uncertainty resolves and performance improves.** Once the firm is fully privatized, the uncertainty about future state exits and the ambiguous mix of objectives is resolved. In addition, privatized firms often experience improvements in competitiveness and productivity, reducing expected default risk. The net effect can be *lower spreads than in the partially privatized state*, even though the guarantee is gone.

7.4 Takeaway

Reducing government ownership raises corporate bond spreads by weakening implicit guarantees, but fully privatized firms borrow more cheaply than partially privatized firms because incomplete and slow privatization creates an intermediate regime of uncertainty and stakeholder conflict that is particularly costly for debtholders.